



0.3.2 Best algorithms of the 20th century ¶

An article published in SIAM News, a publication of the Society for Industrial and Applied Mathematics, lists the ten most important algorithms of the 20th century [\[10\]](#):

1. *1946*: John von Neumann, Stan Ulam, and Nick Metropolis, all at the Los Alamos Scientific Laboratory, cook up the *Metropolis algorithm*, also known as the Monte Carlo method.
2. *1947*: George Dantzig, at the RAND Corporation, creates the *simplex method for linear programming*.
3. *1950*: Magnus Hestenes, Eduard Stiefel, and Cornelius Lanczos, all from the Institute for Numerical Analysis at the National Bureau of Standards, initiate the development of *Krylov subspace iteration methods*.
4. *1951*: Alston Householder of Oak Ridge National Laboratory formalizes the *decompositional approach to matrix computations*.
5. *1957*: John Backus leads a team at IBM in developing the *Fortran optimizing compiler*.
6. *1959–61*: J.G.F. Francis of Ferranti Ltd., London, finds a stable method for computing eigenvalues, known as the *QR algorithm*.
7. *1962*: Tony Hoare of Elliott Brothers, Ltd., London, presents *Quicksort*.
8. *1965*: James Cooley of the IBM T.J. Watson Research Center and John Tukey of Princeton University and AT&T Bell Laboratories unveil the *fast Fourier transform*.
9. *1977*: Helaman Ferguson and Rodney Forcade of Brigham Young University advance an *integer relation detection algorithm*.
10. *1987*: Leslie Greengard and Vladimir Rokhlin of Yale University invent the *fast multipole algorithm*.

Of these, we will explicitly cover three: the decomposition method to matrix computations, Krylov subspace methods, and the QR algorithm. Although not explicitly covered, your understanding of numerical linear algebra will also be a first step towards understanding some of the other numerical algorithms listed.